

Appendix 17

Density functional perturbation theory

In this appendix, we supply a few more details about the Görling–Levy (GL) perturbation theory (Görling and Levy, 1994), i.e., the perturbation theory at constant density. In particular, we provide the explicit expression for the second-order correlation energy $E_c^{(2)}[n]$, defined in Eq. (7.52).

To set up the GL perturbation theory, observe that the external potential $V_\lambda(\vec{r})$, which yields a given density $n(\vec{r})$ at coupling constant λ , can be written as

$$V_\lambda(\vec{r}) = -\frac{\delta F_\lambda[n]}{\delta n(\vec{r})}. \quad (\text{A17.1})$$

The $\lambda = 0$ potential, $V_0(\vec{r})$, yields $n(\vec{r})$ in the non-interacting system, and must therefore be identified with the Kohn–Sham potential: $V_0(\vec{r}) \equiv V_{KS}(\vec{r})$.

We now partition the full hamiltonian $\hat{H}(\lambda)$ of Eq. (7.29) in the following manner:

$$\begin{aligned} \hat{H}(\lambda) &= \hat{H}(0) + \lambda \hat{H}_{e-e} - \sum_{i=1}^N \left(\frac{\delta F_\lambda[n]}{\delta n(\vec{r}_i)} - \frac{\delta F_0[n]}{\delta n(\vec{r}_i)} \right) \\ &= \hat{H}(0) + \lambda \hat{H}_{e-e} - \sum_{i=1}^N [\lambda V_H(\vec{r}_i) + \lambda V_x(\vec{r}_i) + V_{c,\lambda}(\vec{r}_i)], \end{aligned} \quad (\text{A17.2})$$

where $\hat{H}(0)$ is just the Kohn–Sham hamiltonian, and $V_{c,\lambda}(\vec{r}) = \frac{\delta E_{c,\lambda}[n]}{\delta n(\vec{r})}$ is the correlation potential at density n and coupling constant λ . We see that, because the “zero-order” hamiltonian already includes some interaction effects through the Kohn–Sham potential, the “perturbation” consists of the electron–electron interaction *minus* a sum of one-body counterterms. Now from Eq. (7.54) we have

$$V_{c,\lambda}(\vec{r}) = \lambda^2 \frac{\delta E_c^{(2)}[n]}{\delta n(\vec{r})} + \lambda^3 \frac{\delta E_c^{(3)}[n]}{\delta n(\vec{r})} + \dots \quad (\text{A17.3})$$

This enables us to express $\hat{H}(\lambda)$ in a form that is convenient for perturbation theory, namely

$$\hat{H}(\lambda) = \hat{H}(0) + \lambda \hat{H}^{(1)} + \lambda^2 \hat{H}^{(2)} + \dots \quad (\text{A17.4})$$

where

$$\hat{H}^{(1)} = \hat{H}_{e-e} - \sum_{i=1}^N [V_H(\vec{r}_i) + V_x(\vec{r}_i)] , \quad (\text{A17.5})$$

and

$$\hat{H}^{(j)} = - \sum_{i=1}^N \frac{\delta E_c^{(j)}[n]}{\delta n(\vec{r}_i)} , \quad (j \geq 2) . \quad (\text{A17.6})$$

Thus, denoting by $|\psi_k(\lambda)\rangle$ the k -th eigenstate of $\hat{H}(\lambda)$ and by $E_k(\lambda)$ its eigenvalue (with $k = 0$ for the ground-state), we can immediately write down the familiar perturbation expansion for $|\psi_0(\lambda)\rangle$ and $E_0(\lambda)$ in powers of λ :

$$|\psi_0(\lambda)\rangle = |\psi_0(0)\rangle + \lambda \sum_{k \neq 0} \frac{\langle \psi_k(0) | \hat{H}^{(1)} | \psi_0(0) \rangle}{E_0(0) - E_k(0)} |\psi_k(0)\rangle + \dots \quad (\text{A17.7})$$

and

$$E_0(\lambda) = E_0(0) + \lambda^2 \left[\sum_{k \neq 0} \frac{|\langle \psi_k(0) | \hat{H}^{(1)} | \psi_0(0) \rangle|^2}{E_0(0) - E_k(0)} + \langle \psi_0(0) | \hat{H}^{(2)} | \psi_0(0) \rangle \right] + \dots \quad (\text{A17.8})$$

All the eigenfunctions and eigenvalues on the right-hand side of these expressions are obtained from the solution of the Kohn–Sham equation for the ground-state density.

Notice that the first-order term in the expansion of $E_0(\lambda)$ is absent because

$$\langle \psi_0(0) | \hat{H}^{(1)} | \psi_0(0) \rangle = 0.$$

To calculate the second-order term in the perturbative expansion of the correlation energy functional (Eq. (7.54)), i.e., $\lambda^2 E_c^{(2)}$, we differentiate both sides of Eq. (7.51) with respect to λ . Making use of the Hellmann–Feynman theorem we obtain, for small λ ,

$$2\lambda E_c^{(2)} \simeq \langle \psi_0(\lambda) | \hat{H}^{(1)} | \psi_0(\lambda) \rangle - \langle \psi_0(0) | \hat{H}^{(1)} | \psi_0(0) \rangle. \quad (\text{A17.9})$$

Keeping only the first-order term in the expansion (A17.7) of $\psi_0(\lambda)$ we arrive at

$$E_c^{(2)} = \sum_{k \neq 0} \frac{|\langle \psi_k(0) | \hat{H}^{(1)} | \psi_0(0) \rangle|^2}{E_0(0) - E_k(0)} , \quad (\text{A17.10})$$

which is always negative. The usefulness of this result lies in the fact that in small size systems (e.g., atoms) the electronic density is usually very high, so the leading approximation to the correlation energy is often sufficient. For example, Fig. 7.6 shows very clearly that $\frac{dE_{xc,\lambda}}{d\lambda}$ at the ground-state density of a helium atom is an approximately linear function of λ , implying that $E_{c,\lambda}$ is approximately quadratic. Notice, however that the GL expansion of the correlation energy is to be used in conjunction with the DFT definition of exchange (Eq. (7.35)) rather than with the Hartree–Fock exchange. A perturbative expression for the correlation energy, which can be added to the Hartree–Fock exchange is also available and can be found in the paper by Görling and Levy (1994).